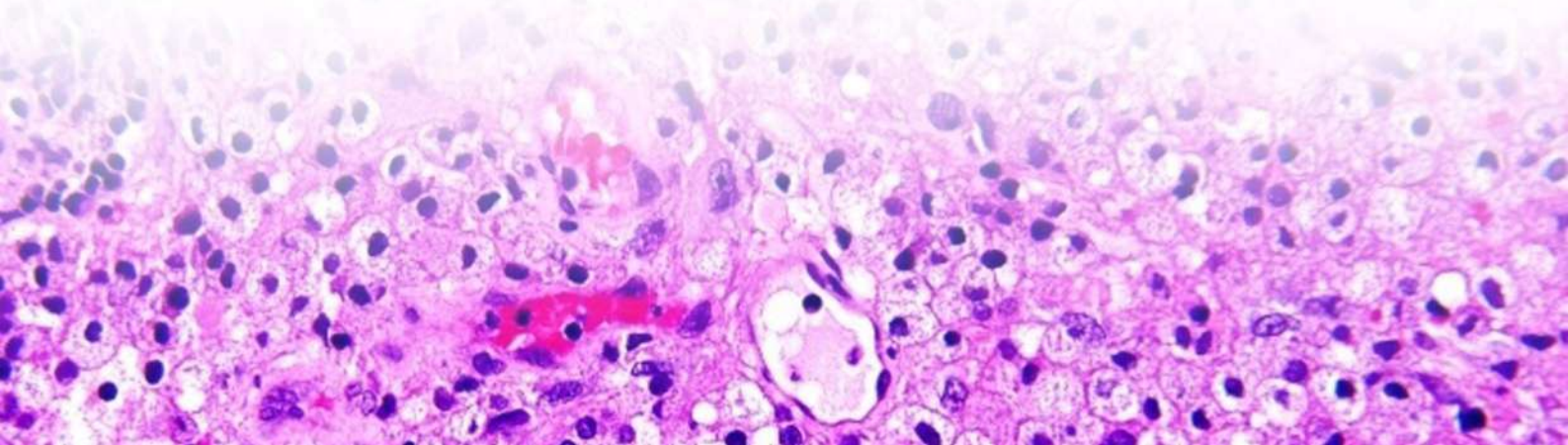
Student Notes

PATHOLOGY

1st Edition

Unit 17

Liver, Gall Bladder, Pancreas



Preface

Agam is a group of budding medicos, who are currently doing their under graduation in various Medical Colleges across Tamil Nadu and Pondicherry. The group was initiated on 18th November 2017, in the vision of uniting medicos for various social and professional causes.

We feel delighted to present you Agam Pathology notes prepared by Agam Divide and Rule 2020 Team to guide our fellow medicos to prepare for university examinations.

This is a reference work of 2017 batch medical students from various colleges. The team took effort to refer many books and make them into simple notes. We are not the authors of the following work. The images used in the documents are not copyrighted by us and is obtained from various sources.

Dear readers, we request you to use this material as a reference note, or revision note, or recall notes. Please do not learn the topics for the 1st time from this material, as this contain just the required points, for revision.

Acknowledgement

On behalf of the team, Agam would like to thank all the doctors who taught us Pathology. Agam would like to whole heartedly appreciate and thank everyone who contributed towards the making of this material. A special thanks to Vignesh M, who took the responsibility of leading the team. The following are the name list of the team who worked together, to bring out the material in good form.

- Neelavathi S
- Fredrick Snowin William
- Subhashree B
- Swathikrishna S
- Sweta Laxmi S
- Varsha L
- Vignesh J
- Vikram R R
- Vimal K
- Habeeb Nathira
- Vennmadhi Vellentina D
- Joy Christy A
- Kirubakaran T
- Swathika N G
- Afrah Marzook
- Dhyaneshwar Ra
- Vimala Veeraiyan
- Kirubakaran T
- Nirupama Sriram
- Subhashree Karishma Choudhury
- Amrutha Priya Devi B
- Vignesh. M

LIVER, GALL BLADDER AND PANCREAS

ESSAY:

1. Classify cirrhosis. Details about Portal cirrhosis.
2. Tumors of Liver.
3. Etiopathology of Viral Hepatitis. Details about serologic evaluation.

SHORT NOTES:

1. Chronic vs acute hepatitis.
2. Metabolic cirrhosis
3. Pathophysiology of Fatty liver
4. Amoebic liver abscess
5. Gall stones
6. Hemochromatosis.
7. Chronic calcific pancreatitis.
8. Acute pancreatitis.
9. LDL.
10. Wilson Disease

SHORT ANSWERS:

1. Conditions of hereditary unconjugated hyperbilirubinemia.
2. Mallory bodies
3. Post Mortem changes of cirrhosis of liver

UPDATES

ESSAY

1. CIRRHOSIS

End stage of chronic liver disease characterised by:

- Entire liver architecture damage
- Bridging fibrous septa
- Regenerating parenchymal nodules
- Variable degrees of vascular (often portosystemic) shunting.

CLASSIFICATION: BASED ON SIZE OF THE NODULES

MICRONODULAR **[NODULE SIZE- LESS THAN 3mm IN DIAMETER]**	MACRONODULAR **[NODULE SIZE- MORE THAN 3mm IN DIAMETER]**
1) Early stage alcoholic cirrhosis	1) Late stage alcoholic cirrhosis
2) Indian childhood cirrhosis	2) Post- necrotic cirrhosis (Viral hepatitis)
3) Hemochromatosis	3) Wilson's disease
4) Primary biliary cirrhosis (PBC)	
5) Jejunoileal bypass	
6) Venous outflow obstruction (Budd-chiari syndrome)	

CLASSIFICATION: BASED ON ETIOLOGY (Mnemonic: $A^2B^2CDEH^2WI$)

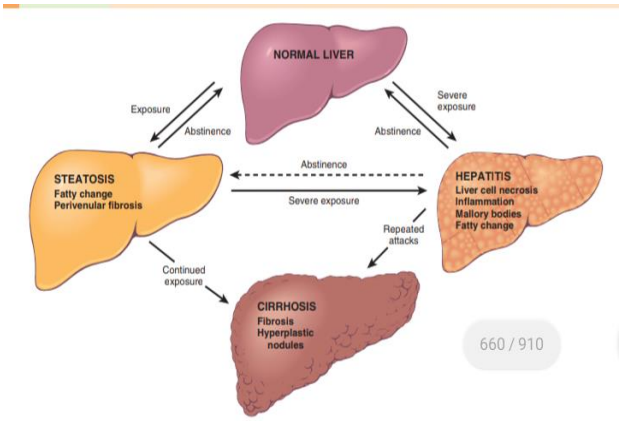
1. Alcoholic Cirrhosis
2. Autoimmune liver Disease
3. Biliary disease (PBC,PSC)
4. Biliary obstruction
5. Cryptogenic cirrhosis; Eg: Non-alcoholic steatohepatitis (NASH)
6. Drug induced and toxins
7. Enzyme deficiency (Alpha1antitrypsin deficiency)
8. Hepatitis
9. Hemochromatosis
10. Wilson's disease
11. Indian Childhood cirrhosis

PORTAL CIRRHOSIS:

- Also known as **Laennec's cirrhosis, alcoholic cirrhosis, nutritional cirrhosis**.
- Common cause- chronic alcohol consumption.

ALCOHOLIC LIVER DISEASE:

- Excessive alcohol consumption- 80gm or more ethanol causes significant liver injury.
- Consists of three distinct forms:
  - ✓ **Hepatocellular Steatosis or fatty change**
  - ✓ **Alcoholic (-steato) hepatitis**
  - ✓ **Fibrosis and cirrhosis**



PATHOGENESIS:

- Alcohol —NADH→ Acetaldehyde —NADH→ Acetate
  Alcohol dehydrogenase **Aldehyde dehydrogenase**
- **Hepatic Steatosis**:
  - **Accumulation of intracellular lipids.**
  - NADH used for lipid biosynthesis and ethanol impairs assembly of lipoproteins.

- **Alcoholic hepatitis:**
  - Acetaldehyde- induces lipid peroxidation, disrupt cytoskeleton and membrane function.
  - Alcohol- affects mitochondrial function and membrane fluidity.
  - ROS- Kupffer cell activation, oxidative damage to hepatocyte membrane.
- **Cirrhosis:** Pericellular and sinusoidal fibrosis first develops, connecting portal triad with central veins forming regenerative nodules.
- Loss of hepatocyte and liver shrinks.

MORPHOLOGY:

- **STEATOSIS:**
  - Initially **microvesicular** type and later **macrovesicular** type.
  - Start at **Centrilobular** area and become panlobular.

- **ALCOHOLIC HEPATITIS:**
  - **Hepatocyte ballooning (**hepatocyte swelling)
  - **Mallory-Denk bodies(**intracytoplasmic hyaline inclusion, Cytokeratin 8 and 18 intermediate filaments)
  - Neutrophilic infiltrations
  - Centrilobular necrosis
- **STEATOFIBROSIS:**
  - **Central vein sclerosis**
  - **Chicken wire fence pattern**(perisinusoidal scarring spreads outward,encircling small clusters of hepatocyte)
  - **Central portal fibrous septa**
  - Irreversible stage- initially micronodular and later macronodular
  - Entire liver lobe is replaced by tough pale scar tissue.

GROSS: Liver is yellow-tan, fatty and enlarged, with persistent damage liver turns into brown shrunken, non-fatty organ composed of cirrhotic nodules.

CLINICAL FEATURES:

Symptoms and laboratory abnormalities range from minimal to severe.

- Malaise, anorexia, weight loss, upper abdominal discomfort, nausea and vomiting, tender hepatomegaly and fever.
- Development of jaundice and encephalopathy.
- Scleral icterus, palmar erythema, spider angioma, parotid gland swelling, ascites, edema, clubbing and muscle wasting.
- Male- decreased body hair, Gynecomastia, testicular atrophy.
- Women- menstrual irregularities.

LABORATORY TESTS:

- **Serum total bilirubin elevated**
- Direct bilirubin mildly elevated
- **Neutrophilic leukocytosis**
- Prolonged prothrombin time
- **Serum AST and ALT elevated**
- Deficiency of vitamin B12, thiamine.

COMPLICATIONS:

- Hepatic failure
- Pancreatitis
- Massive gastrointestinal haemorrhage
- Intercurrent infection
- Hepatorenal syndrome
- Hepatocellular carcinoma

2. TUMORS OF LIVER

The liver is the most common site of metastatic cancers from primary tumors of colon, lungs and breast.

BENIGN TUMORS:

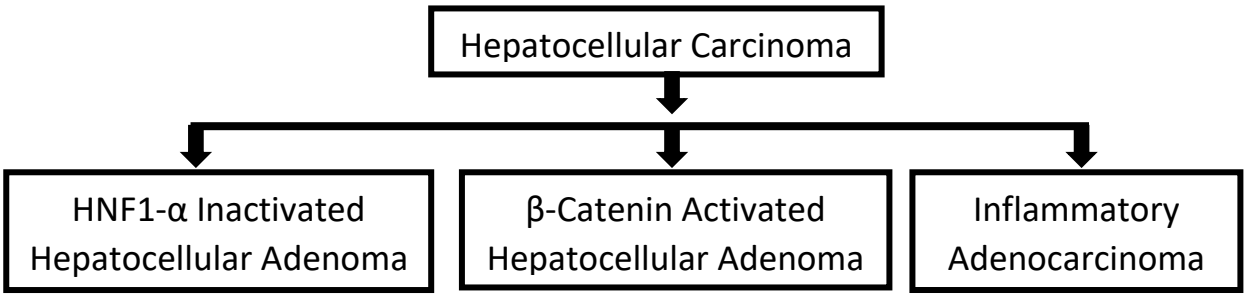
A. CAVERNOUS HEMANGIOMA

- Most common benign tumor of liver.
- It is composed of large,dialated vascular channels filled with blood.
- They do not spontaneously regress.
- Difficult to distinguish from malignant tumor.
- **Morphology:**
  - Red blue, soft nodules, usually less than 2 cm in diameter.
  - Located beneath the capsule**.**
- **Histological examination:** It consist of vascular channel in the bed of connective tissue.

B. HEPATOCELLULAR ADENOMA

- Benign neoplasms from Hepatocytes
- Detected – incidentally
  - After abdominal pain(due to pressure by growing tumour on the capsule)
  - After haemorrhagic necrosis
- May rupture → Intraabdominal bleeding → Surgical importance
- Risk factors – Oral contraceptives, Anabolic Steroids

Based on molecular analysis, HCA is of 3 types:



TYPE	**PATHOGENESIS**	MORPHOLOGY
HNF1-α Inactivated Hepatocellular Adenoma	• 90% - Inactive somatic mutation of HNF1-α • 10% - Genetic Mutation • HNF1-α codes for a transcription factor • Commonly found in women	• Tumors are fatty and devoid of cellular or architectural atypia • Liver Fatty Acid Binding Protein(LFABP) is absent in tumor cells (expressed in normal cells) • Thus, immune staining for LFABP is diagnostic
β-Catenin Activated Hepatocellular Adenoma	• Activating mutations of β-catenin • Have a high risk for Malignant transformation • Should be resurrected even when asymptomatic	• Have high degree of cytological or architectural Dysplasia • Immuno staining for β-Catenin shows nuclear translocation indicative of its activated sites • Glutamate synthase, a target of β-Catenin is also positive (usually around perivenular hepatocytes
Inflammatory Hepatocellular Adenoma	• Activating mutations of gp130, a coreceptor for IL-6, which lead to JAK-STAT signalling and overexpression of cute phase reactants • Associated with Non-Alcoholic Fatty Liver Disease • 10% of IHA have concomitent β-Catenin Activated mutation	• Unlike other HCAs, these lesions have areas of fibrotic stroma, mononuclear inflammation, ductular reactions, dilated sinusoids and telangiectatic vessels • Express Acute Phase Reactants like CRP

MALIGNANT TUMOR

- **Hepatocellular Carcinoma -**Most common malignant tumor
- **Hepatoblastoma- M/C** malignant tumor in Children
- **Angiosarcoma- M/C** Sarcoma

HCC:

- Primary liver cancer arise from hepatocytes.
- Cirrhosis is not required for hepatocarcinogenesis.
- Western country-90% HCC develop into cirrhotic liver.
- Asian country-50% HCC develop into non cirrhotic liver.
- Accounts for 5.4% of cancers worldwide.
- 85% of cases are associated with chronic HBV infection.
- Peak incidence: 20 – 40 years of age.
- It is also called hepatoma
- **High risk Areas**
  - Asia.
  - Africa.
  - Increasing incidents in USA.

ETIOLOGY:

- Chronic liver disease are most common setting for emergence of HCC
- Etiological factors-
  - viral infections : HBV,HCV
  - Toxins: aflatoxin(mycotoxin) produced By aspergillus
- Other risk factors:
  - Alcohol
  - Cigarette smoking
  - Inherited disorders:
    - ✓ alpha AT deficiency
    - ✓ Wilson's disease
    - ✓ Hemochromatosis
  - Metabolic disorders: obesity, DM, NAFLD

PATHOGENESIS:

- HCC is induced by acquired driver mutations in oncogenes and tumor suppressor genes.
- Gain of mutations in beta-canenin.
- Loss of mutations in P53.
- More recent studies implicate a role or signalling through the IL-6/JAK/STAT pathway in HCC. IL-6 is overproduced in many chronic hepatitides.
- IL-6 can suppress hepatocyte differentiation and promote their proliferation by regulatory the function of transcription factor HNF4-α.

- **Premalignant precursors of HCC**
  - Hepatic Adenoma
  - Chronic liver disease associated with cellular Dysplasia
  - Low grade dysplastic nodules
  - High grade dysplastic nodules (viral hepatitis and Alcohol liver disease)

MORPHOLOGY:

- Masses appears grossly as unifocal / multifocal, diffuse.
- Intrahepatic metastasis occurs through vascular invasion and direct extension.
- Extra hepatic metastasis by vascular invasion through portal vein and inferior venacava.
- **Color:**
  - White (abundant stroma),
  - Yellow (predominant fatty changes)
  - Green (malignant hepatocytes secrete bile).

HISTOLOGY:

- Large pseudo acinar space
- Malformed dilated bile canaliculi
- Thickened hepatocyte trabeculae

DIAGNOSIS:

- 50% cases -elevated serum alpha feto protein level
- Ultrasonography, computed tomography, magnetic resonance imaging
- Mass will progressively increase

CLINICAL FEATURES:

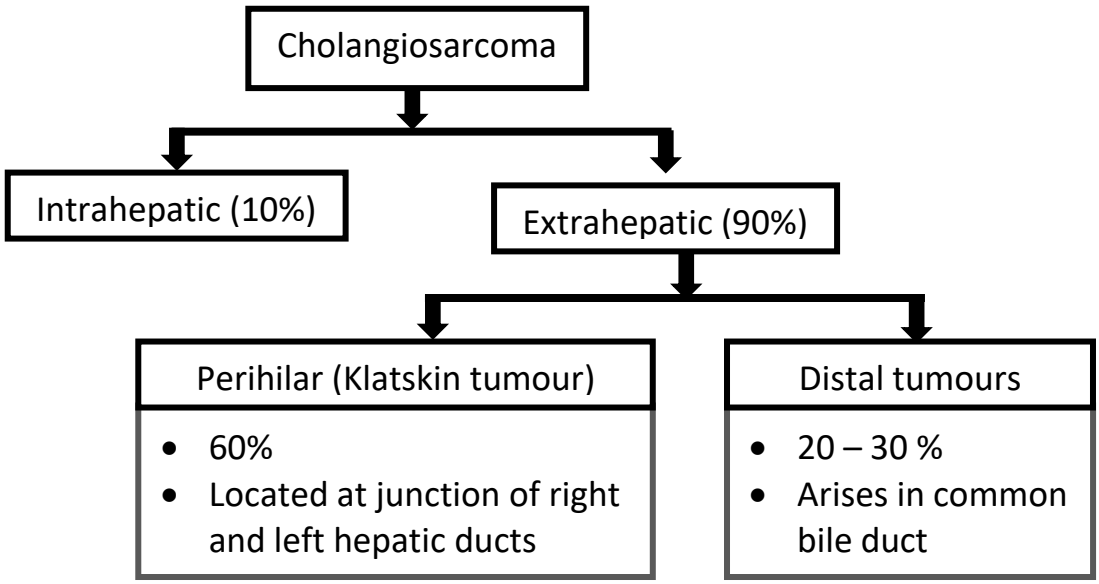
- Upper abdominal pain, malaise, fatigue, weight
- Other inconstant findings-jaundice, fever, gastrointestinal or oesophageal varices
- Death occur due to cachexia, liver failure with hepatic coma, rupture of tumor with fatal hemorrhage, gastrointestinal or oesophageal varices
- Majority of patients dies within 2years of diagnosis

TREATMENT:

- Small tumors removed surgically or through embolisation, microwave radiation and freezing.
- Advanced stage -liver transplantation.
- Life can be prolonged by Sorafenib(kinase inhibitor).

CHOLANGIOSARCOMA (CHOLANGIOCARCINOMA)

- Second most common primary tumour inn liver next to HCC.
- It is the malignancy of biliary tree within or outside the liver.
- More common in areas with liver fluke is endemic.
- **Risk factors**:
  - ✓ Infestation by liver fluke
  - ✓ Primary sclerotic cholangitis
  - ✓ Hepatolithiasis
  - ✓ Fibropolycystic liver disease
  - ✓ HBV, HCV
  - ✓ Non-alcoholic fatty liver



CLINICAL FEATURES:

- **Intrahepatic:** Usually asymptomatic during early course of the tumour. They become detectable only in their late course. Hence, median time from diagnosis to death is only 6 months even after surgery.
- **Extrahepatic;** present with symptoms of biliary obstruction, cholangitis and right upper quadrant pain. Median time from diagnosis to death is 2 years.

MORPHOLOGY:

- Cholangiosarcomas, regardless of the site, are Adenosarcomas and secrete **Mucin**.
- Most are differentiated with clearly defined glandular/tubular structures
- Extrahepatic CSs:
  - Generally small when diagnosed as they cause rapid obstructive symptoms
  - **Firm, gray / diffusely infiltrative/ papillary polypoid.**
- Intrahepatic CSs:
  - Occur in non cirrhotic liver
  - Track along the intrahepatic portal tract system.

3. ETIOPATHOLOGY - VIRAL HEPATITIS & SEROLOGICAL EVALUATION:

VIRAL HEPATITIS:

Defined as viral infection of hepatocyte that produces **bridging** necrosis and inflammation.

BRIDGING NECROSIS

- Bridging necrosis is a more severe form of hepato-cellular injury in acute viral hepatitis and may progress to fulminant hepatitis or chronic hepatitis.
- In bridging necrosis, the necrotic zone may **link central veins to portal tracts** or bridge adjacent portal tracts (often with an inapparent central vein within the zone of injury).

CAUSES: It is caused by hepatotropic viruses such as hepatitis A, B,C,D,E viruses.

INVESTIGATIONS OF VIRAL HEPATITIS:

COMMON CONSIDERATIONS:

- Total bilirubin /urine bilirubin: Elevated.
- Liver panel / serum aminotransferases: Elevated.
- Prothrombin time: If prolonged indicates severe impairment of liver function.
- Blood urea nitrogen (BUN) and serum creatinine: Decreased renal function indicates Fulminant hepatic disease.
- Mild normochromic normocytic anaemia with thrombocytopenia and leukopenia is present even before portal hypertension and splenomegaly.
- Demonstration of virus in faeces, blood, bile by immunoelectron microscopy.
- Detection of antibody by ELISA and molecular diagnosis by RT-PCR.
- Serum ammonia: if raised then hepatic encephalopathy.

HEPATITIS A VIRUS:

- Usually benign condition.
- Viral genome: Non enveloped single stranded RNA.
- Receptor: HAVcr-1(membrane glycoprotein).
- Transmission-feco-oral route and close personal contact with.
- Incubation period-2 -6 weeks.
- Affected individual develop nonspecific symptoms (fatigue, loss of appetite) and jaundice.
- It will not cause chronic hepatitis.
- HAV mediated Hepatocellular injury caused by CD8+ T cells.
- HAV is shed in stool (2-3 weeks before and 1 week after onset of jaundice).
- It can be also detected in serum and saliva.

SEROLOGICAL EVALUATION:

- Detection of serum Ig M.
- IgM response usually decline after few months followed by appearance of IgG anti HAV (no test for Ig G anti HAV-(total anti HAV-IgM anti HAV)

HEPATITIS B VIRUS

- **Antigen**: Hepatitis B virus
- **Viral Family:** Hepadnavirus
- **Genome:** Partially ds circular DNA with 4 open reading frames which code for markers -
  - Core Region : **HBcAg** (Hepatitis B core antigen) - participates in the assembly of complete virions in hepatocytes
  - Precore Region : **HBeAg** (Hepatitis B e antigen) - directs the secretion of the HBeAg polypeptide into blood
  - Envelope glycoprotein : **HBsAg** (Hepatitis B surface antigen) - synthesised and secreted by infected hepatocytes
  - DNA Polymerase: **Pol** -DNA polymerase and reverse transcriptase activity.
  - **HBx Protein**: Virus replication and pathogenesis of HBV associated liver carcinoma.

TRANSMISSION:

- High prevalence region – perinatal transmission (during childbirth)
- Intermediate prevalence region – horizontal transmission (common in children)
- Low prevalence region – unprotected sex, IV drug abuse

HBV INFECTION :

- Incubation period – 2 to 26 weeks.
- Outcomes – The outcome is determined by host immune response
- Innate immunity acts during initial phase.
  - Acute hepatitis followed by recovery and clearance of the virus.
  - Non progressive chronic hepatitis.
  - Progressive chronic disease ending in cirrhosis.
  - Fulminant hepatitis with massive liver necrosis.

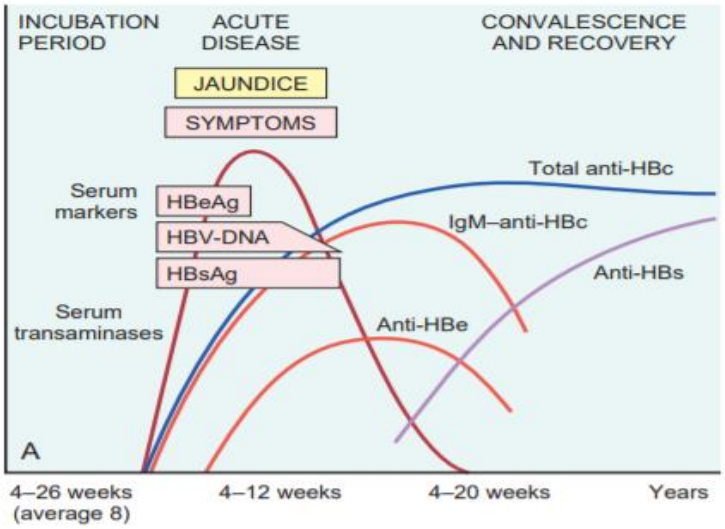
ACUTE INFECTION WITH RESOLUTION:

CLINICAL MANIFESTATION:

- Subclinical or anicteric hepatitis.
- Icteric hepatitis
- Fulminant hepatitis.

Nonspecific Symptoms: Right upper quadrant discomfort, nausea, jaundice, anorexia, fever.

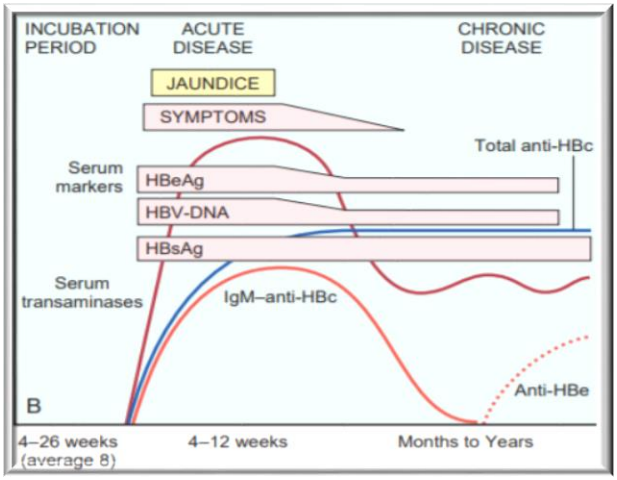
SEROLOGIC MARKERS:



- **HBsAg** - appears before onset of symptoms, Peaks during overt disease, Declines within 12-24 weeks.
- **HBeAg , HBV-DNA, DNA pol** – appears after HBsAg, Indicates continued viral replication, Persistence of HBeAg is an indicator of progression to chronic hepatitis
- **Anti-HBs Antibody** – appears after HBsAg declines, May persist for life and confer protection – rationale for current vaccines containing HbsAg.
- **IgM-Anti-HBc** – detectable shortly before the onset of symptoms, Concurrent with onset of elevated ALT/AST, IgM replaced by IgG in due course of time
- **Anti-HBe Antibody** – signifies acute infection peak has been reached and will continue to subside.

Resolution of acute infection is associated with CD4+ and CD8+ IFN – gamma producing cells (hepatocyte injury is caused by CD8+ cytotoxic T cells attacking infected cells).

CHRONIC INFECTION



GROUND GLASS HEPATOCYTE: Cases of **chronic hepatitis B** show scattered Ground-glass hepatocytes indicative of abundance of HBsAg in the cytoplasm. Endoplasmic reticulum of hepatocyte is swollen by the antigens. It is a diagnostic hallmark of HBV infection.

- The progression rate from acute to chronic HBV infection decreases with age
- Complete resolution is not possible as the virus inserts itself in the host DNA.
- Leads to impairs development of immune response.

CLINICAL MANIFESTATION:

Asymptomatic carrier state to chronic hepatitis, cirrhosis, and hepatocellular carcinoma. Advanced liver disease there may be clinical signs of chronic liver disease including **splenomegaly, spider angioma, caput medusae, palmar erythema, testicular atrophy, gynecomastia.**

HEPATITIS C VIRUS:

- It is a small enveloped ssRNA.
- Etiological factors-intravenous drug abuse, multiple sex patterns, surgery within last 6 months, needle stick injury, employment in medical field
- In children vertical perinatal transmission present
- It has viral proteases for processing NS5A(protein for assembly)and RNA polymerase
- Incubation period-4-26 weeks
- More often it causes chronic hepatitis.

SEROLOGICAL EVALUATION:

Serum aminotransferases : The values do not predict the severity of clinical findings, the degree of histologic abnormalities, the patient's prognosis, or the therapeutic response. Normalization of AST and ALT levels after acute infection may signal clearance of HCV.

SEROLOGIC ASSAYS :

- **Anti-HCV test :**
  - Recombinant HCV antigens are used in enzyme-linked immunosorbent assay (ELISA) to detect IgG anti-HCV in patients' sera.
  - Can remain negative even after months of acute infection.
  - Anti-HCV is not a protective antibody.
- **Recombinant immunoblot assays (RIBAs):** More specific.
- **HCV RNA testing.**

HEPATITIS D VIRUS:

- It is a circular defective ssRNA.
- Delta agent depends on HBV for it's life cycle.
- Superinfection of HBV carrier by HDV has two phase.
- **Acute phase**
  - HDV replicate by suppression of HBV
  - ALT level is high.
- **Chronic phase**
  - HBV replicate by suppression of HDV
  - ALT level fluctuates

SEROLOGICAL EVALUATION:

- Detection of HDV RNA in Blood and liver at onset of acute symptomatic disease.
- Coinfection of HBV and HDV-IgM against HD AG and HBcAg is detected.
- HDV superinfection-HBsAg and anti HDV antibody (IgM and IgG).

HEPATITIS E VIRUS:

- It is ssRNA.
- Enterically transmitted waterborne self limited disease
- Zoonotic disease-epidemic
- Cause 30-60% acute hepatitis
- High mortality among pregnant woman
- Incubation period-4-5 weeks
- Undeveloped RNA virus in the hepavirus genus.

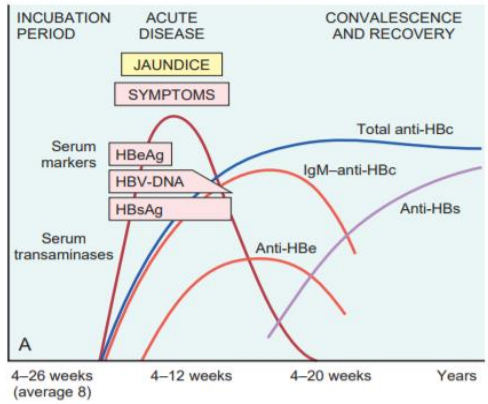
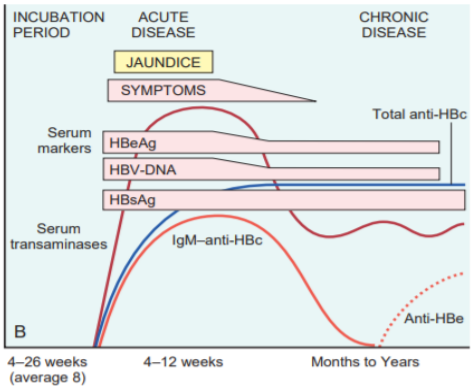
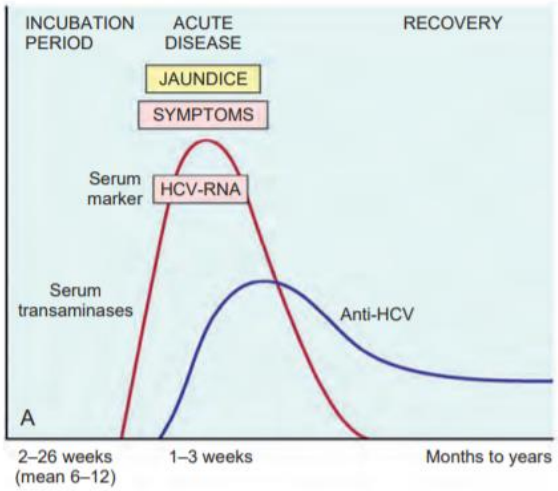
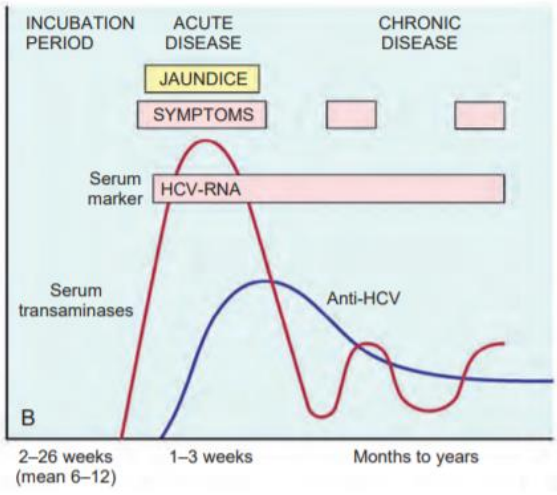
SEROLOGICAL EVALUATION:

- HEV RNA and HEV virion detected by PCR in stools and serum.
- Rising serum aminotransferase.
- Elevated IgM anti HEV and latter IgG anti HEV.

SHORT NOTES:

1. ACUTE VS CHRONIC:

ACUTE	CHRONIC
Histological findings	
▪ Gross: Normal or slightly mottled and greenish. Might shrunk under acute liver failure. ▪ Microscopically: Mononuclear infiltrate Portal inflammation: absent or minimal Hepatocellular injury - ballooning degeneration, councilman body or acidophil body, dropout necrosis, bridging necrosis Spotty necrosis or Lobular hepatitis ▪ Severe Acute hepatitis : confluent necrosis around central vein	▪ Mononuclear portal inflammation. ▪ Interface Hepatitis (At the interface between hepatocellular parenchyma and portal tract stroma) ▪ Scarring and Fibrosis. ▪ Microscopically: Piecemeal necrosis, portal tract lesions, intralobular lesion's, bridging necrosis
HBV	
▪ Subclinical or anicteric hepatitis. ▪ Icteric hepatitis. ▪ Fulminant hepatitis. ▪ Nonspecific Symptoms**:** Right upper quadrant discomfort, nausea, jaundice, anorexia, fever.	▪ Asymptomatic or healthy carriers ▪ In advanced Cases: splenomegaly, spider angioma, caput medusae, palmar erythema, testicular atrophy, and gynecomastia. ▪ In decompensated cirrhosis, jaundice, ascites, peripheral edema, and encephalopathy may be present ▪ Ground glass hepatocytes with ER swollen with HBsAg (diagnostic hallmark) is a histological finding

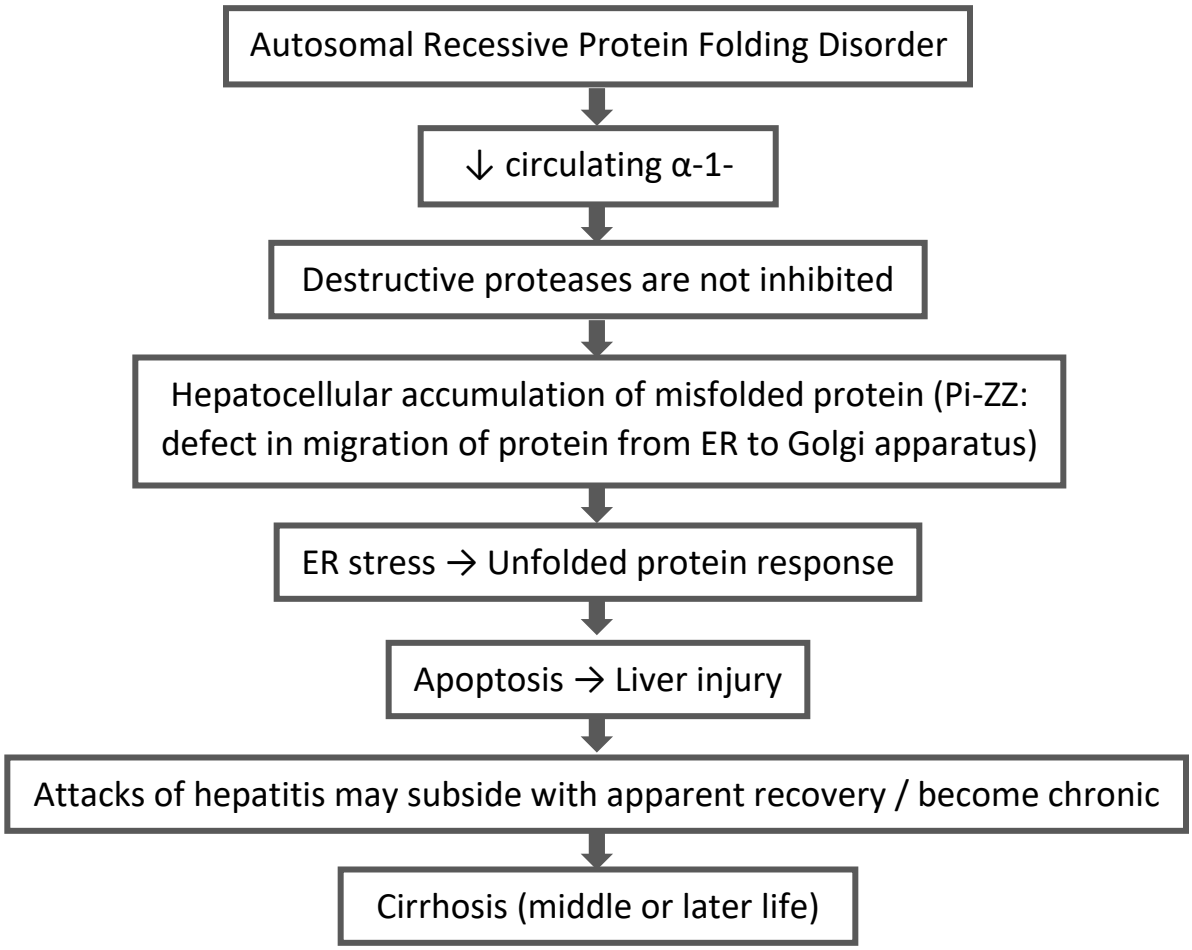
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<td>More common in adults.</td>
<td>Advancement to chronic is more common in Children.</td>
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<tr>
<td colspan="2"><u>HCV:</u></td>
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<tr>
<td>▪ Asymptomatic.
▪ Symptomatic (milder than HBV infection)</td>
<td>▪ Asymptomatic.
▪ Severe cases cirrhosis.
▪ Persistent elevation in serum transaminases (wax and wane)
▪ gives rise to insulin resistance and NAFLD (met syndrome)
▪ Lymphoid aggregates and Fatty change are histological findings.</td>
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2. METABOLIC CIRRHOSIS:

A number of metabolic conditions involving the liver can cause chronic liver disease leading to cirrhosis.

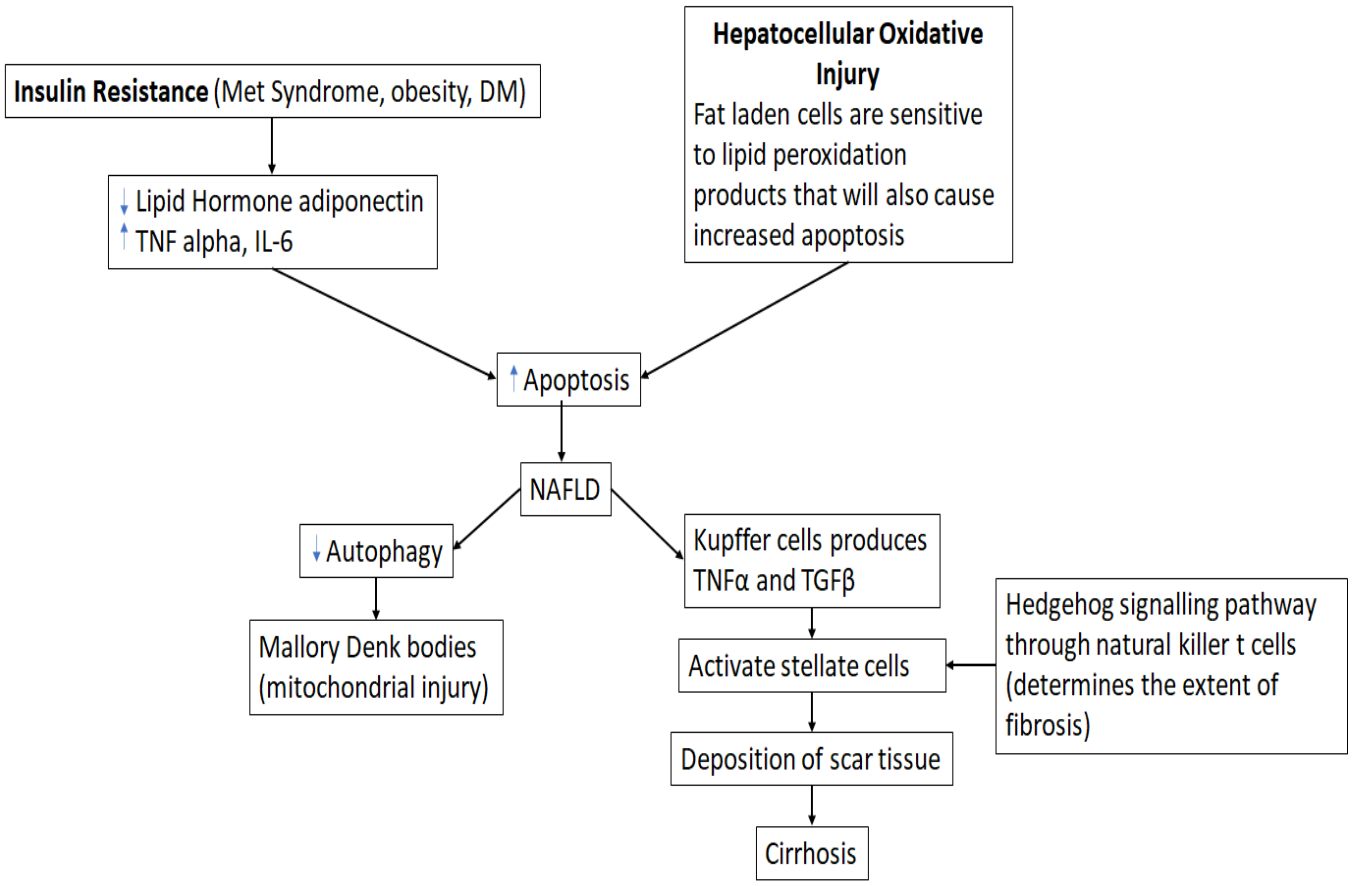
CIRRHOSIS: Diffuse transformation of the entire liver into regenerative parenchymal nodules surrounded by fibrous bands and variable degrees of vascular shunting.

ALPHA1-ANTITRYPSIN DEFICIENCY:

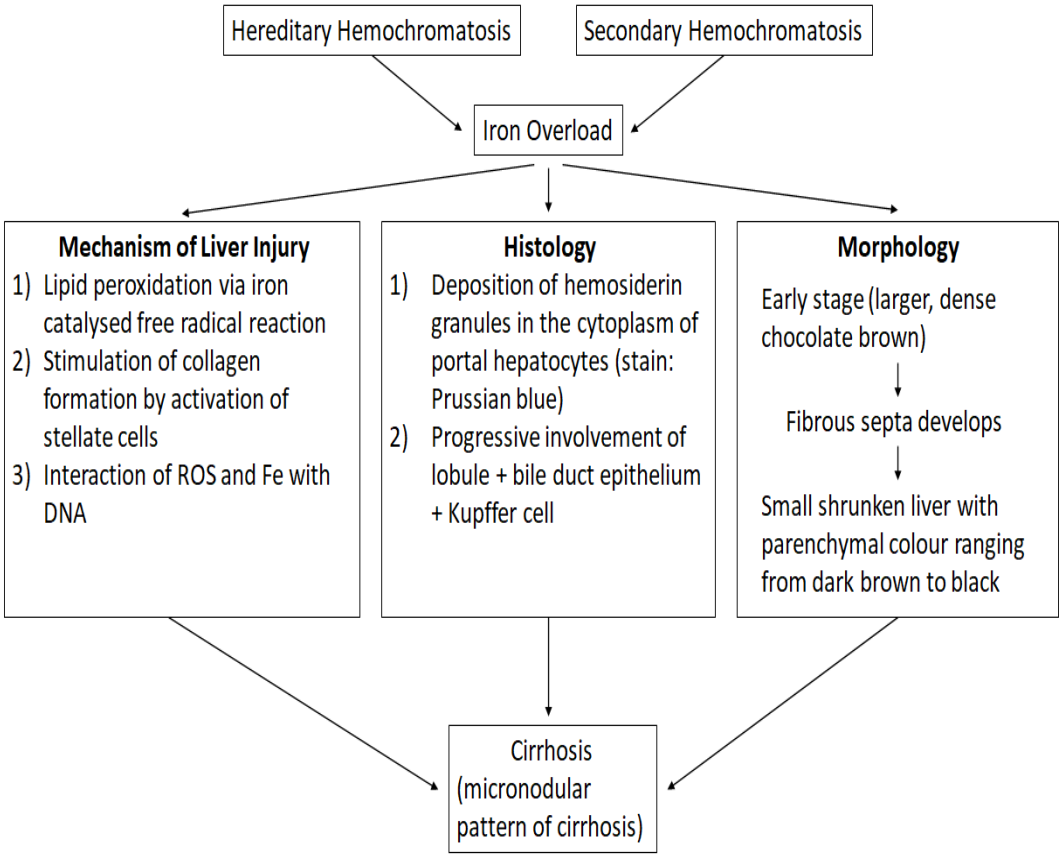


GENOTYPE	ALPHA1-AT LEVELS
Pi-MM	Normal
Pi-ZZ	10% of normal
Pi-MZ	Intermediate

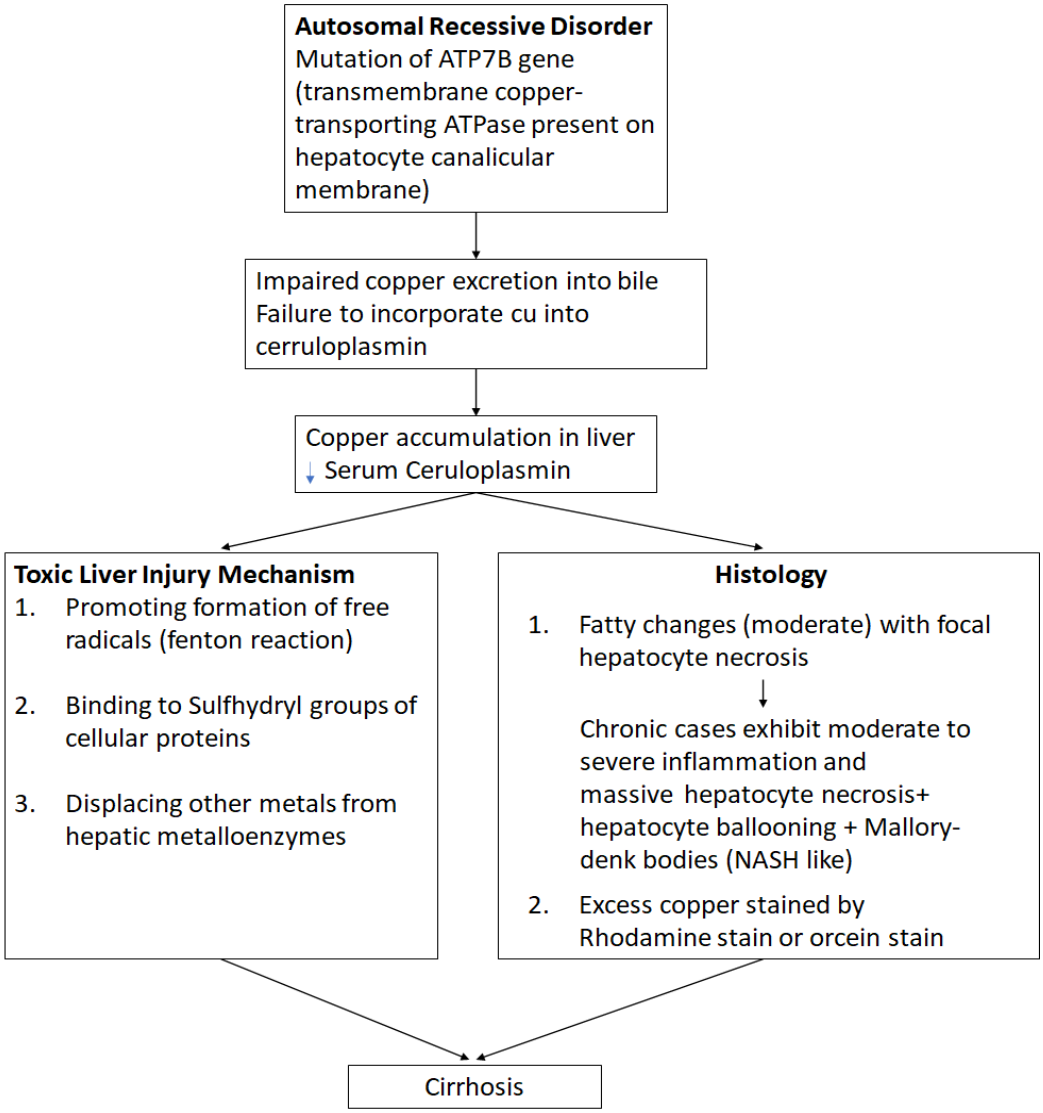
NON ALCOHOLIC FATTY LIVER DISEASE:



HEMOCHROMATOSIS



WILSON DISEASE



3. PATHOPHYSIOLOGY - FATTY LIVER

Fatty change in liver may results from one of the two types of causes

CONDITIONS WITH EXCESS FAT: Conditions in which capacity of the liver to metabolise fat is exceeded:

- Obesity, Diabetes mellitus.
- Congenital hyperlipidaemia.

LIVER CELL DAMAGE: Conditions in which fat cannot be metabolized due to liver cell injury:

- Alcoholic liver disease (most common)
- Starvation.
- Protein calorie malnutrition.
- Chronic illnesses (e.g. tuberculosis)
- Acute fatty liver in late pregnancy
- Hypoxia (e.g. anaemia, cardiac failure)
- Hepatotoxins (e.g. carbon tetrachloride, chloroform, ether, etc)
- Drug induced liver cell injury (steroids, CCl4, etc)
- Reye's syndrome.

PATHOPHYSIOLOGY – ALCOHOLIC FATTY LIVER

- Gender: Women are more prone to fatty liver than men may be due to alcohol pharmacokinetics and metabolism and estrogen dependent response to gut derived endotoxins (LPS) in liver.
- Estrogen increases gut permeability to endotoxins which in turn increase the expression of LPS receptor CD14 in kupffer cells which increases production of proinflammatory cytokines.
- Ethnic and genetic difference: US cirrhosis higher for African Americans than white American.
- Genetic polymorphism in detoxifying enzymes and some cytokines promoters plays a significant role.
- ALDH*2 variant of aldehyde dehydrogenase unable to oxidise acetaldehyde and do not tolerate alcohol leading to alcohol intolerance- upper body flushing, nausea and lethargy.
- Combined conditions: Iron over load and infection with HCV and HBV synergize alcohol leading to severity of liver disease.

Hepatocellular steatosis results from:

- Shunting of normal substrates towards lipid biosynthesis resulting in increased NADH.
- Impaired assembly and secretion of lipoprotein.
- Increased peripheral catabolism of fat releasing free fatty acids into circulation.
- Acetaldehyde intermediate metabolite of alcohol induces lipid peroxidation and acetaldehyde protein adduct formation and dysrupts cytoskeleton membrane function.
- Alcohol impairs metabolism of methionine and decrease glutathione, and thereby sensitizing liver to oxidative injury.
- Converts drugs like acetaminophen to toxic metabolites (by induction of cytochrome P450 enzyme, which causes alcoholic catabolism in endoplasmic reticulum).
- Release of bacterial endotoxins into portal circulation induce inflammatory response by NF-kB and TNF, IL-6.
- Alcohol also stimulates release of endothelins from endothelial cells causing vasoconstricton and contraction of activated myfibroblastic cell leading to decrease in hepatic sinusoidal perfusion.

NON-ALCOHOLIC FATTY LIVER DISEASE:

- Insulin resistance leads to hepato steatosis
- Hepatocellular oxidative injury leads to liver cell necrosis and inflammatory reactions.
- Hepatic steatosis like obesity occurs due to overabundance of calorie rich food, diminished exercise and epigenetics modifications.
- In insulin resistance and metabolic syndrome visceral adipose tissue becomes dysfunctional leads to reduced production of lipid hormone, adiponectin and increased inflammatory cytokines such as TNF and IL-6 promoting hepatocyte apoptosis.
- Diminished autophagy leads to mitochondrial injury and formation of Mallory-Denk bodies.
- Kupffer cells produce TNF and TGF activating the stellate cells and leading to scar formation.
- Stellate cell is activated through hedgehog pathway thus, stages of fibrosis correlates to the level of hedgehog pathway.

4. HEPATIC AMOEBIASIS:

- It's a common extra intestinal complication.
- Amoebic dysentery absent in 50 percent
- Patients with amoebic colitis develop a tender liver without impairment of liver function
- It might be because of repeated invasion of amebae from colonic infection or toxic substance reach the liver, causing damage (lysosomal enzymes of lysed polymorphonuclear neutrophils and monocytes and cytokines from inflammatory cells surrounding the trophozites).

AMOEBIC LIVER ABSCESS:

- It is a type of liver abscess caused by amebiasis.
- It is the involvement of liver tissue by trophozoites of the organism **Entamoeba histolytica** and of its abscess due to necrosis.
- **Grossly :**
  - The center of abscess-thick chocolate brown pus (anchovy sauce pus) Liquified necrotic liver tissue.
  - Periphery-normal tissue contains invading amoeba.
  - It might be multiple or solitary usually in the upper right lobe (poesteriosuperior portion)
- **Histologically:**
  - The necrotic area consists of degenerated liver cells, leucocytes, red blood cells, strands of connective tissue and debris.
  - Amoebae are most easily found in the liver tissue at the margin of abscess.
- **PAS-staining** is employed to confirm the trophozoites of **Entamoeba histolytica**.

CLINICAL FEATURES:

- Painful hepatomegaly, fever, anorexia, nausea.
- Weight loss and fatigue.
- Leukocytosis , increased serum transaminases, jaundice develops when they obstruct biliary tract.
- Untreated abscess penetrate into diaphragm into lung or pleural cavity,pericardium,peritoneal cavity, stomach and IVC.

5. GALL STONES:

RISK FACTORS:

- **Age and sex**- middle to older age
  - Prevalence high in female, hyper secretion of biliary cholesterol plays the role in age and sex difference.
  - Risk increases with sex and metabolic syndrome.
- **Environmental Factors:** Estrogen exposure- oral contraceptive and pregnancy increase expression of hepatic lipoprotein receptor and stimulate HMG-CoA activity and enhance cholesterol uptake and biliary secretion
- **Acquired Disorder:** Gall bladder stasis either neurogenic or hormonal favorable for gall stones.
- **Hereditary Factor:** Genes encoding hepatocyte proteins that transport biliary lipids known as ATP- binding cassette (ABC). The sterol transporter encoded by ABCG8 is also involved.
- **Demographic** – Western world.
- **High calorie diet and obesity.**
- **Drugs** – clofibrate, exogenous estrogens.
- **Gastrointestinal disorders.**
- **Pigment gallstones** – disorders associated with erythrocytes, cirrhosis of any cause, and infection of the biliary tract by liver flukes.

PATHOGENESIS OF CHOLESTROL STONES: (crystalline cholesterol monohydrate)

Cholesterol concentration exceeds the solubilizing capacity of the bile.

- Super saturation of cholesterol with bile
- Hypo motility of gall bladder
- Accelerated cholesterol crystal nucleation
- Hyper secretion of mucus in gall bladder trapping nucleated Crystal's

PATHOGENESIS OF PIGMENT STONES: (bilirubin calcium salts)

- Complex mixtures of inorganic calcium salts of unconjugated bilirubin.
- Common in elevated unconjugated bilirubin- chronic hemolytic anemia (secretion of conjugated bilirubin increases lots of deconjugated bilirubin to pigment stones).
- Severe ileal dysfunction.
- Bacterial contamination of biliary tree- release of microbial glucoronidases which hydrolyze bilirubin glucuronides.

MORPHOLOGY:

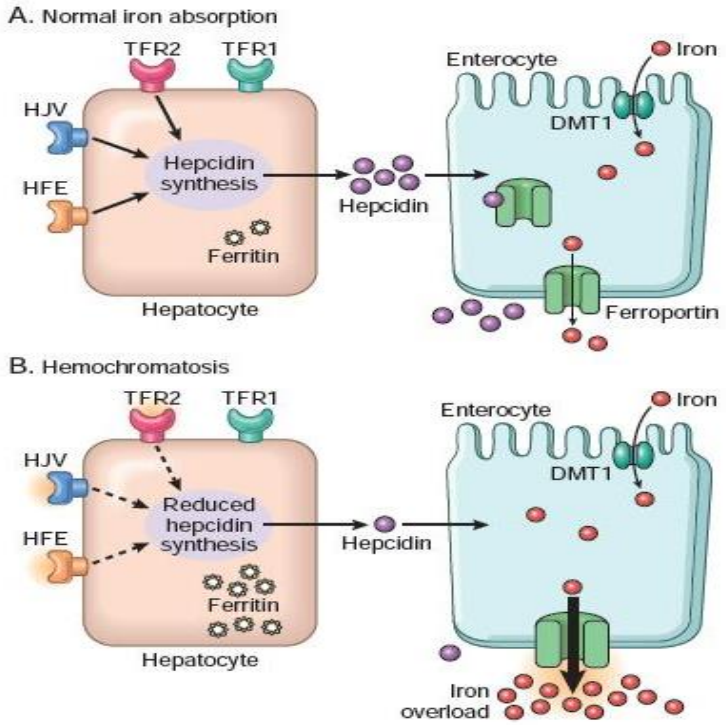
- **Cholesterol stone:**
  - Pale yellow, round to ovoid, fine granular, hard external surface
  - Transection-glistening radiating crystal palisade.
  - Multiple stones are present, may be rounded or faceted because of tight apposition.
  - Mostly they are radiolucent, stones with sufficient calcium carbonate are radiopaque.
- **Pigment gall stones:**
  - Brown- found in sterile gall bladder bile.
  - Black- infected large bile ducts.
  - Contours are spiculated and molded.
  - Brown stones are:
    - ✓ Laminated
    - ✓ Soft-soap like or greasy
    - ✓ Radiolucent
  - Black stones are: radiopaque.

CLINICAL FEATURES: PATHOLOGICAL EFFECTS OF GALL STONE:

- Asymptomatic
- Biliary colic after a fatty meal
- Pain localized to right upper quadrant or right epigastric radiating to right shoulder or back
- Complications: empyema, perforations, fistulas, inflammation of the biliary tree (cholangitis), obstructive cholestastis and pancreatitis.
- Large stones erode in small bowel causing intestinal obstruction -gall stone ileus or **Bouveret syndrome.**
- Increased risk of gall bladder carcinoma

6. HEMOCHROMATOSIS

- Caused by excessive iron absorption.
- Liver; pancreas; heart; joints; endocrine glands.
- **3 types:**
  - **Hereditary hemochromatosis**: mutations of genes encoding HFE; TfR2; HJV
  - **Secondary hemochromatosis/hemosiderosis**: transfusions; sideroblastic anemia; thalassemia; chronic liver disease
  - **Neonatal hemochromatosis/congenital hemochromatosis**: manifested by severe liver disease and extrahepatic hemosiderin deposition.
- **Triad of features:**
  - Micro nodular pigment cirrhosis.
  - Diabetes mellitus.
  - Skin pigmentation.



BRONZE DIABETES:

Pathogenesis:

- HFE / HJV / TFR2 gene mutations.
- Reduced hepcidin synthesis – diminished circulating hepcidin.
- Decreased hepcidin-ferroportin interaction.
- Increased ferroportin activity.
- Increased iron efflux from enterocytes – systemic iron overload.

Morphology:

- Deposition of hemosiderin {liver > pancreas > myocardium > pituitary gland > adrenal gland > joints > skin} – appear as **golden yellow pigment** granules in the cytoplasm of affected parenchymal cells.
- Cirrhosis.
- Pancreatic fibrosis.

Clinical features:

- Hepatomegaly.
- Abdominal pain.
- Abnormal skin pigmentation
- Deranged glucose homeostasis / Diabetes mellitus.
- Cardiac dysfunction (arrhythmias, cardiomyopathy).
- Atypical arthritis.

7. CHRONIC CALCIFIC PANCREATITIS

Prolonged inflammation of pancreas with the presence of stones in it and does not heal or improve.

- **Etiology:**
  - Heavy alcohol abuse.
  - Autoimmune conditions.
  - Genetic mutation due to cystic fibrosis.
  - Blocked pancreatic duct / common bile duct.
  - Familial pancreatitis.
- **Pathophysiology:**
  - Mislocation of cystic fibrosis transmembrane conductance regulator chloride channel in pancreatic duct cells.
  - Decrease in pancreatic HCO3 secretion.
  - Low pH of pancreatic juice leads to fluid mal secretion.
  - Protein plug formation in the pancreatic ducts.
  - CALCIFICATION – Pancreatic stone formation.

MORPHOLOGY:

- **Gross appearance:**
  - Pancreas – enlarged, firm, nodular.
  - Cut section – smooth grey appearance with loss of normal lobulation .
  - Foci of calcification and tiny pancreatic concentrations to large visible stones – Present.
- **Microscopic examination:**
  - Obstruction of the ducts by fibrosis in the wall and protein plugs or stones in the Lumina.
  - Chronic inflammatory infiltrate around the lobules and ducts.
  - Atrophy of acinar tissue.
- **Diagnosis:** Visualization of calcifications within the pancreas by CT or Ultrasonography.

CLINICAL FEATURES:

- Epigastric abdominal pain which radiates to back.
- Pancreatic insufficiency {malabsorption with steatorrhea and fat-soluble vitamin deficiencies}.
- Dystrophic calcification of pancreatic parenchyma on imaging.
- Weight loss and edema.

8. ACUTE PANCREATITS

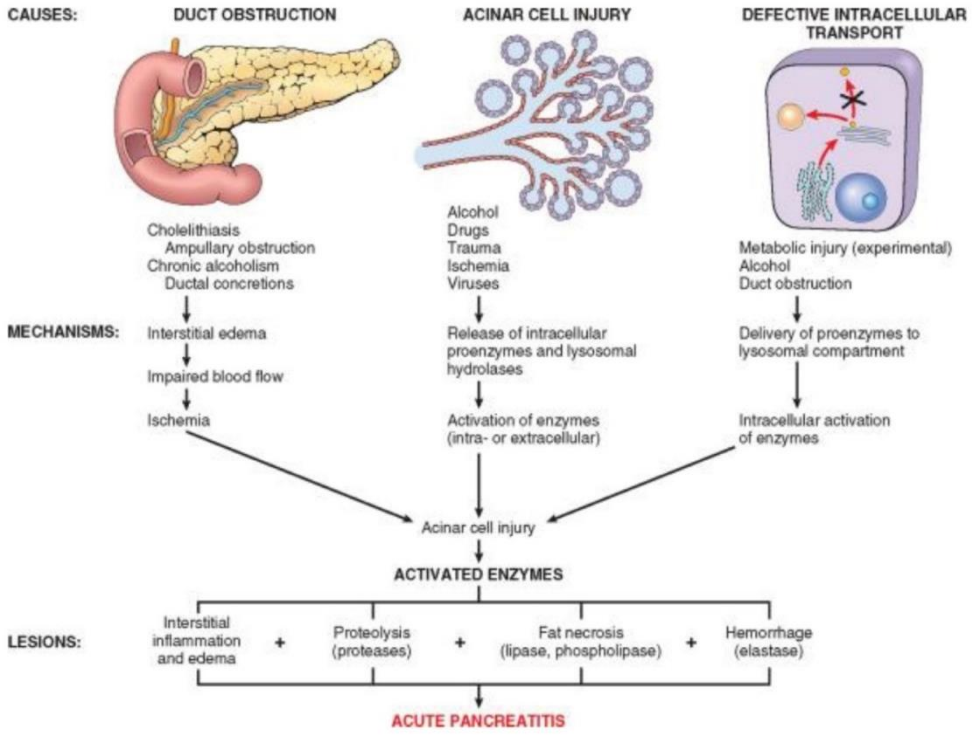
- Characterized by reversible pancreatic parenchymal injury associated with inflammation.
- Acute hemorrhagic pancreatitis / Acute pancreatitic necrosis.

ETIOLOGY:

- Alcoholism: Gene Polymorphism for UDP (Uridine-5-Diphosphate) Glucuronyl Transferase → Increase risk for Alcohol Induced Pancreatitis.
- Certain drugs (e.g. azathioprine, thiazide, sulfonamides, oral contraceptives).
- Mutation in cationic trypsinogen (PRSS1) & Trypsin inhibitor (SPINK1) genes
- Extension of inflammation from the adjacent tissues.
- Hypercalcaemia from hyperparathyroidism.
- Blood-borne bacterial infection.
- Cholelithiasis.
- Trauma.
- Ischaemia.
- Shock.
- Hypothermia.
- Hyperlipoproteinemia.
- Mumps

PATHOGENESIS:

- Destructive changes in the pancreas are attributed to the liberation and activation of pancreatic enzymes – proteases, lipases and phospholipases, elastases
- Activation and release of this enzyme is brought about by:
  - Acinar cell damage
  - Duct obstructions
  - Block in exocytosis



GROSS APPEARANCE:

- Pancreas (early stage) – swollen, edematous.
- Variegated appearance of grey-white pancreatic necrosis..
- Chalky – white fat necrosis.
- Blue – black hemorrhages are seen

MICROSCOPIC EXAMINATION:

- Necrosis of pancreatic lobules and ducts.
- Necrosis of arteries and arterioles with area of hemorrhages.
- Fat necrosis.

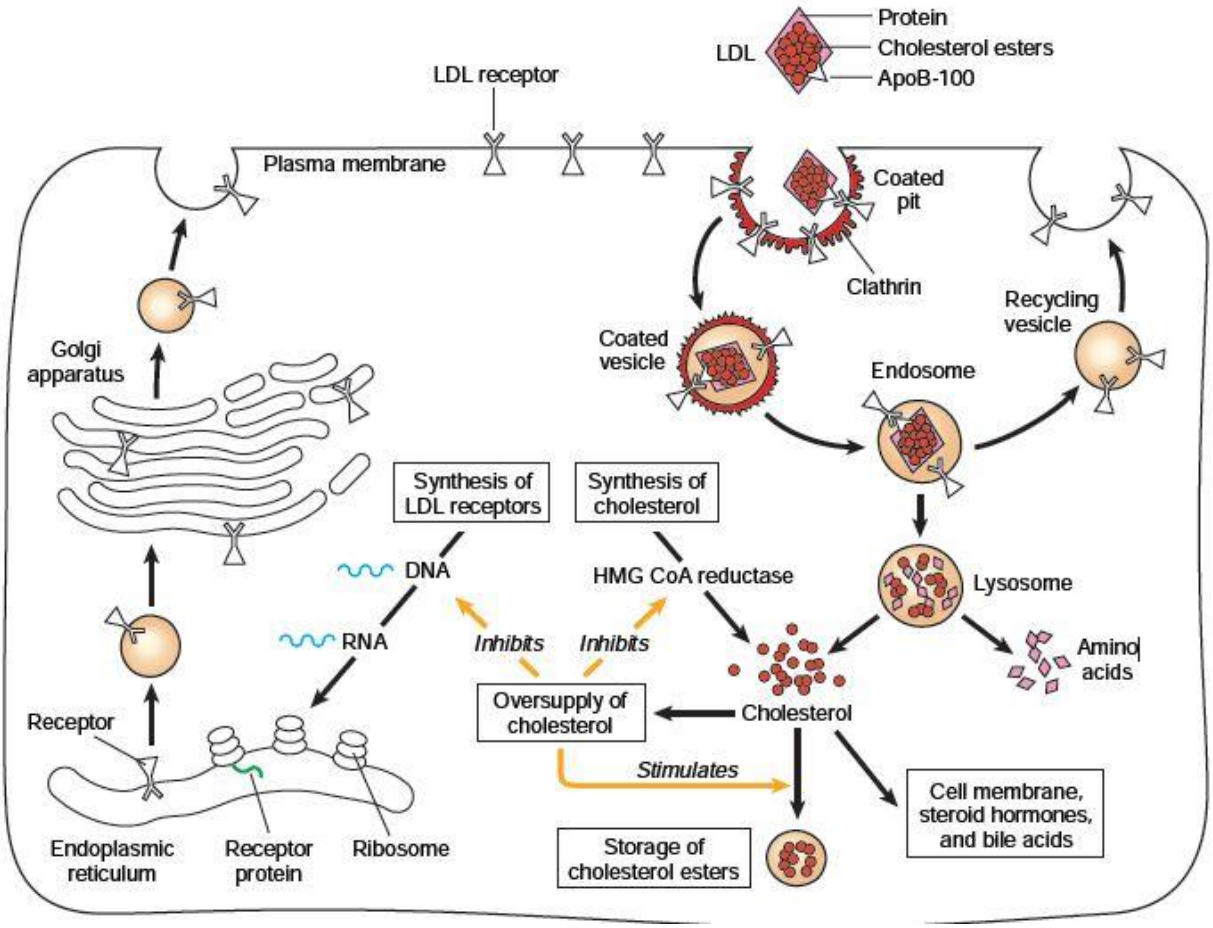
COMPLICATIONS:

- **Systemic complications:**
  - Chemical and bacterial peritonitis.
  - Endotoxic shock.
  - Acute renal injury.
- **Local complications:**
  - Pancreatic abscess.
  - Duodenal obstruction.

9. LDL:

- LDL stands for Low-Density Lipoprotein.
- Nearly 7% of body's cholesterol circulate in the plasma, predominantly in form of LDL.

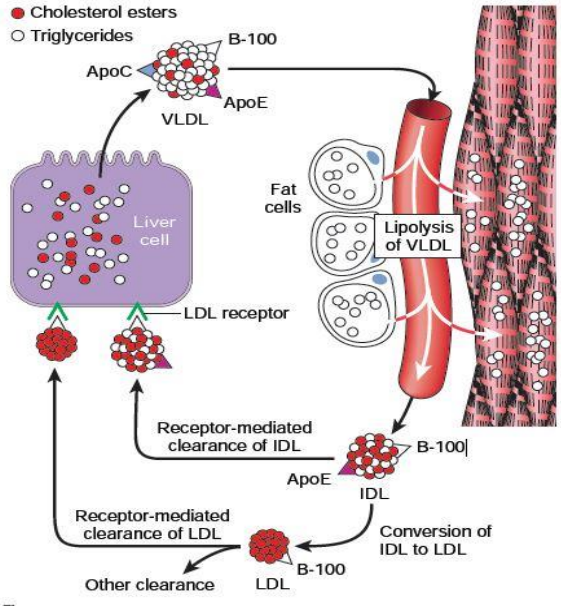
LDL RECEPTOR PATHWAY AND REGULATION OF CHOLESTEROL METABOLISM:



FAMILIAL HYPERCHOLESTEROLEMIA:

- Is is a receptor disease that results from mutations encodingthe receptor for LDL.
- Since, the LDL receptor levels are reduced, catabolism of LDL is impaired leading to increase in plasma cholesterol level.
- IDL also uses LDL receptors and hence high levels of IDL are diverted to synthesis of even more LDL.
- Elevated seum cholesterol levels increases the risk of ischemic heart diseases, atherosclerosis and other coronary artery diseases.
- More than 900 mutations are found responsible for Famiilial Hypercholesterolemia and are classified into 5 classes:

Class I : Failure of synthesis of receptors.
Class II : Folding defects in ER leading to impaired transport of proteins synthesised in ER to Golgi bodies.
ClassIII : Defect in LDL binding domain in the receptor. Receptor reach the surface of cell, but LDL will not be able to bind to its receptors.
Class IV : Failure in formation of coated pits. LDL bind to receptor, but fails to internalise.
Class V : pH dependant dissociation of receptor is affected. LDL gets inernalized but fail to get recycled.



10. WILSON'S DISEASE / HEPATOLENTICULAR DEGENERATION

Causes & etiology:

- Autosomal Recessive.
- Mutation of **ATP7B Gene** (chromosome 13) which encodes transmembrane copper transporting ATPase protein.
- Mutation results in **impaired in copper excretion** into bile & failure to incorporate copper into Ceruloplasmin.

Pathogenesis:

- Normally in Liver copper binds to alpha 2 Globulin (Apo ceruloplasmin) to form Ceruloplasmin, which is sent into blood. It is affected in Wilson's disease.
- Accumulated Copper causes Toxic Liver injury by 3 mechanisms
    - Promoting the Formation of Free Radicals by FENTON REACTION
    - Binding to Sulfhydryl Groups of Cellular Protein
    - Displacing other metals from Hepatic metalloenzymes

Morphology:

- FATTY CHANGE (**Steatosis**)
    - May be mild to moderate with Focal hepatocyte Necrosis
    - Acute Fulminant Hepatitis mimic Acute Viral Hepatitis
    - Chronic Hepatitis in Wilson's Exhibits – Inflammation, Hepatocytes Necrosis, Fatty Change, Features of Steatohepatitis (Mallory-Denk bodies)
- Toxic injury to the brain primarily affects the basal ganglia, particularly the putamen, which shows atrophy and even cavitation.

Clinical features:

- Liver: cirrhosis, liver failure.
- Brain: movement disorders, dementia, and psychiatric issues.
- Kidney: renal disease.
- Eye: **Kayser–Fleischer's ring**, sunflower cataract.
- Blood: Hemolytic anemia.

Diagnosis:

- ↓Serum Ceruloplasmin.
- ↑ Hepatic copper content (most Sensitive & Accurate)
- ↑ Urinary excretion of Copper (most Specific).

Treatment: Copper Chelation therapy with **D Penicillamine/Trientine**, Zinc Based therapy.

SHORT ANSWERS:

1. HEREDITARY UNCONJUGATED HYPERBILIRUBINEMIA

DISORDER	INHERITANCE	DEFECT	CLINICAL FEATURES
Criggler-Najjar Syndrome 1	**AR**	Absence of UGT1A1 activity	**Fatal in neonatal period.**
Criggler-Najjar Syndrome 2	**AD**	Decreased UGT1A1 activity	**Generally mild, Occasional Kernicterus.**
Gilbert Syndrome	**AR**	Decreased UGT1A1 activity	**Jaundice during Stress otherwise, Innocuous.**

2. MALLORY BODIES:

- Mallory Bodies are usually present as clumped, amorphous, eosinophilic material in ballooned hepatocytes.
- They are made up of tangled skeins of intermediate filaments such as keratins 8 and 18 in complex with other proteins such as ubiquitin.
- Seen in: (MNEMONIC: New Indian WATCH)
  - Non-alcoholic fatty liver disease
  - Indian Childhood Cirrhosis
  - Wilson's disease
  - Alpha1anti trypsin deficiency
  - Tumors of liver
  - Cholestasis
  - Hepatic/biliary disease

3. POST-MORTEM CHANGES OF CIRRHOSIS OF LIVER.

- Softened and flabby-12 to 24 hours in summer
- Multiple blisters- 24 to 36 hours
- **Cl.Welchii** form characteristic small clumps in tissue space and produce gas
- Small, opaque, yellowish-grey, dendritic figures in parenchyma
- **Honeycombed appearance**
- **Putrefaction crystals** (composed of tyrosine and leucine) on surface of liver
- Colour changes:
  - **Deep blue**- passive congestion
  - Green-oozing of bile from gall bladder
  - **Blue-black**- in subcapsular area, pigment from adjacent loop of intestine
  - Green a few hours after- jaundice, pigment oxidised to biliverdin
- In advanced putrefaction- liver becomes green or black

UPDATES

1. SINUSOIDAL CAPILLARISATION:

- If the injury and inflammatory stimuli persist, ECM deposition and scarring commence, often in the space of Disse; there is concurrent loss of sinusoidal endothelial cell fenestration, a change termed sinusoidal capillarization.
- It is a particularly prominent feature in nonalcoholic steatohepatitis and also is a feature of the abnormal sinusoids seen in hepatocellular carcinoma.

2. INCOMPLETE SEPTAL CIRRHOSIS

- If the chronic injury leading to scar formation is interrupted (e.g., clearance of hepatitis virus infection or cessation of alcohol use), then stellate cell activation and scarring ceases and fibrous septae may begin to be broken down by metalloproteinases produced by hepatocytes, leading to partial resolution and an appearance termed incomplete septal cirrhosis.
- vascular remodeling and other architectural changes that occur in cirrhosis may not revert to normal
- Vascular abnormalities such as portal hypertension fail to improve in some patients.

3. PATHOGENESIS OF ACUTE LIVER FAILURE

- Triggering event for acute liver failure is hypothesized to be portal hypertension and 2o ↑ production of vasodilators such as nitric oxide by endothelial cells in the splanchnic vasculature.
- This in turn leads to systemic vasodilation and diminished renal perfusion, which is sensed by the kidney, provoking activation of the renin/angiotensin axis.
- In the setting of portal hypertension and persistent vasodilator production, the principal effect of renin/ angiotensin activation is to further diminish renal perfusion and glomerular filtration rate, leading to renal failure.
- The decline in renal function is reversible if liver function is restored, for example, by liver transplantation

4. VIRAL HEPATITIS-C PROTEINS

- Two envelope proteins, E1 and E2, and five core proteins: p7, NS2, NS3/4a (protease), NS5A (replication complex), and NS5B (RNA polymerase).

5. AUTOIMMUNE HEPATITIS

- Autoimmune hepatitis has a strong association with specific HLA alleles in Caucasians (DR3), Japanese (DR4), and in South Americans (DRB1).
- Proposed triggers for the immune reaction include viral infections, drugs/toxins, and vaccination

Morphology:

- Extensive inflammation and hepatocellular injury at the interface, as well as in the hepatic parenchyma, is characteristic of autoimmune hepatitis.
- Lymphocytes and plasma cells may be seen within the cytoplasm of hepatocytes, often at the interface, a curious phenomenon called emperipolesis.

Type of necrosis seen:

- confluent necrosis
- Panacinar necrosis
- Bridging necrosis
- "rosettes," a circular arrangement of hepatocytes around a dilated canaliculus

6. MC TYPE OF DRUG INDUCED HEPATITIS-

- Idiosyncratic reactions are the most common form of drug-induced liver injury and usually occur after 1 to 3 months of exposure
- Metabolism of isoniazid (an antituberculous agent) is slow in individuals
- Variant of N-acetyltransferase (NAT2) with decreased enzymatic activity, raising drug levels in the liver and leading to susceptibility to isoniazid hepatotoxicity. Antimicrobial drugs are the most common culprits in idiosyncratic reactions, accounting for nearly one-half of cases
- N-acetyl_P-benzquinone immine (NAPQI), produced by the cytochrome P-450 system. Pericentral zone 3 hepatocytes are most sensitive to NAPQI,

7. LIVER ENZYME ROLE IN DIAGNOSIS OF DRUG INDUCED HEAPATITIS:

Liver enzyme tests can be used to gauge whether the injury is primarily

- Hepatocellular (alanine aminotransferase [ALT] ≥5 times the upper limit of normal, or ALT/alkaline phosphatase [ALP] ratio >5),
- Cholestatic (ALP ≥2 times the upper limit of normal, or ALT/ALP ratio <2), or
- Mixed (increased ALT and ALP with ALT/ALP ratio between 2 and 5)

8. MORPHOLOGY OF DRUG AND TOXIN INCLUDED LIVER INJURY

- Idiosyncratic reactions show typical features of acute hepatitis dominated by inflammation and varying degrees of necrosis.
- centrizonal necrosis is a common feature.
- Injury caused by intrinsic hepatotoxins is dominated by necrosis with minimal inflammation. Injury centered on bile ducts is characterized by varying combinations of cholestasis and ductular reactions, which can progress to chronic cholestasis and duct "dropout" in a minority of cases
- Drugs like- amiodarone (antiarrhythmic), tamoxifen (anti-estrogen), irinotecan (antineoplastic), methotrexate (immunosuppressive) can cause a steatohepatitis-like pattern of injury.
- Mitochondrial dysfunction caused by drugs such as tetracycline (antibiotic), valproic acid (anticonvulsant), zidovudine (antiretroviral) can result in microvesicular steatosis.
- Endothelial injury to sinusoids and central veins can be caused by cytotoxic agents (azathioprine, oxaliplatin), leading to sinusoidal obstruction syndrome.

9. NONALCOHOLIC FATTY LIVER DISEASE-

Pathogenesis-

- Complex associations with genetic variants, diet, and the intestinal microbiome established association between increased gut-derived endotoxin production and liver inflammation and injury.
- High-fructose diets have also been associated with increased risk of NAFLD-related fibrosis, and dietary fat, particularly trans-fat, may have a role in producing liver injury.
- Obstructive sleep apnea, usually occurring in the setting of obesity, has been associated with disease progression, possibly related to intermittent hypoxia.
- Fibrosis may be accelerated when injury from another liver disease
- Reduced production of the hormone adiponectin from adipocytes, which decreases oxidation of free fatty acids by skeletal muscle and increases free fatty acid uptake into hepatocytes
- Hepatocytes in NAFLD down-regulate lipolysis, an alteration that may further contribute to the accumulation of lipids.
- Lipolysis occurs through a mechanism called lipophagy that closely resembles macroautophagy, which you will recall is the process by which cells remove excessive or dysfunctional cellular components and (in times of starvation) generate metabolites for energy production

Morphology

- Steatosis (≥5% of hepatocytes), lobular inflammation, and ballooned hepatocytes are required for its diagnosis.
- Alcoholic hepatitis has on average less steatosis and more ballooned hepatocytes, lobular inflammation, Mallory hyaline, neutrophilic infiltrates, cholestasis, and obliterated central veins.
- Fibrosis typically develops around the central vein as a fine "spider web" of pericellular collagen deposition
- Ballooned hepatocytes may not be present.

10. FATTY LIVER DISEASE

- Hepatic steatosis (fatty liver) is an early, predictable effect of alcohol consumption.
- Macroscopically, the fatty liver of chronic alcoholism is enlarged (as heavy as 4 to 6 kg), soft, yellow, and greasy.
- Macrovesicular steatosis is the predominant form in alcoholic liver disease. An unusual exception is alcoholic foamy degeneration, a form of microvesicular steatosis sometimes seen with chronic heavy alcohol use that is associated with endoplasmic reticulum and mitochondrial damage.
- Ballooned hepatocytes - These are injured, swollen hepatocytes with cleared-out cytoplasm and cytoskeletal damage,
- Mallory hyaline may reflect a failed attempt to sequester and degrade damaged cytoplasmic proteins.

11. NEONATAL CHOLESTASIS

Morphology: Inflammation and fibrosis of the hepatic or common bile ducts is the hallmark of the disease.

Diagnosis-

- The combination of clinical presentation, imaging (ultrasound and hepatospecific iminodiacetic-acid [HIDA] scans), and biopsy can confirm the diagnosis in a vast majority of cases.
- Ultrasound findings of a small or absent gallbladder and fibrosis at the porta hepatis support the diagnosis of extrahepatic biliary atresia, as do HIDA scans with technetium-99m (99mTc), which is excreted in bile.
- In extrahepatic biliary atresia, there is total lack of secretion of 99mTc into the bile, and the biliary tree is not visualized.

13. NONOBSTRUCTIVE NEONATAL CHOLESTASIS-

Causes

- Loss-of-function mutations JAG1 or the receptor NOTCH2, both of which are required for normal development of the biliary tree.
- Certain inborn errors of metabolism, particularly galactosemia& Niemann-Pick disease
- Nonsyndromic causes of nonobstructive cholestasis include α1-antitrypsin deficiency

Morphology

- Decreased numbers of bile ducts within portal regions profiles is the most characteristic and constant feature.
- The most striking features are hepatocanalicular cholestasis and giant-cell or syncytial cell change with multinucleated hepatocytes

14. CHRONIC CHOLECYSTITIS

- **Morphology:** There is complete replacement of the gallbladder wall and mucosa by dense fibrosis (hyalinizing cholecystitis), with or without calcification.
- Other rare forms of chronic cholecystitis include IgG4-related sclerosing cholecystitis, another manifestation of IgG4-related fibrosing disease.

3

15. MUTATION SEEN IN CHOLANGIOCARINOMA

- KRAS mutations are common in both tumors.
- Intrahepatic cholangiocarcinoma have distinctive driver mutations in IDH1 and IDH2 (isocitrate dehydrogenase) that generate “oncometabolites”
- Fusion genes involving FGFR2 (fibroblast growth factor receptor 2) are also common.
- By contrast, in addition to KRAS mutations, extrahepatic biliary adenocarcinomas are most likely to have mutations in TP53 and SMAD4, genetic features that are similar to those of pancreatic adenocarcinoma.

16. PRIMARY SCLEROSING CHOLANGITIS

- The gold standard for diagnosis is the characteristic “beading” seen in the large intrahepatic & extrahepatic biliary tree seen in ERCP/MRCP, involving attributable to irregular biliary strictures and dilations
- Inflammatory bowel disease, particularly ulcerative colitis, affects approximately 70% of individuals with PSC, whereas 8% of inflammatory bowel disease patients develop PSC.
- Mostly in association with inflammatory bowel disease, there is involvement of smaller bile ducts only and the ERCP/MRCP examination is normal. This is referred to as **small-duct PSC** and can progress to typical large-duct PSC.

17. FOCAL NODULAR HYPERPLASIA:

- Focal nodular hyperplasia (FNH) is a benign, non-neoplastic lesion that is most common in adult women.

Pathogenesis-

- Altered blood flow that leads to hyperplastic changes in hepatocytes.
- FNH-like nodules can occur adjacent to other tumors and in vascular conditions such as hemangiomas and Budd-Chiari syndrome.
- Multifocal FNH can occur in a/w hepatic or extrahepatic hemangioma, vascular malformations, or (curiously) brain tumors, such as meningioma and astrocytoma.

Morphology-

- poorly formed elastic lamina
- A prominent ductular, reaction is often present, but interlobular bile ducts are absent.
- The hepatocytes in the nodules are arranged in one- to two cell-thick plates; cytologic and architectural atypia is not present.
- Immunohistochemistry for glutamine synthetase (GS), an enzyme normally present in centrizonal hepatocytes, is very useful for diagnosis.
- In FNH, a highly characteristic “maplike pattern” of strong cytoplasmic GS staining is seen within anastomosing groups of hepatocyte.

Clinical feature-

- Large regenerative nodules can occur in noncirrhotic liver in vascular conditions such as Budd-Chiari syndrome, hemangioma, and vascular malformations.
- In some conditions, the entire liver becomes nodular without significant fibrosis, a condition referred to as nodular regenerative hyperplasia (NRH).
- NRH can lead to portal hypertension and mimic cirrhosis on imaging bile duct adenoma and biliary hamartoma (von Meyenburg complex).
- Both are asymptomatic and generally less than 2 cm
- Both lesions show haphazard bile ducts in a fibrous stroma, and lack atypical cytologic or architectural features.
- The ductular profiles in bile duct adenoma tend to be round and nondilated, while those in biliary hamartoma are dilated, curvilinear and often show inspissated bile or secretions.

BECKWITH WIEDEMANN SYNDROME.

- Caused by abnormal epigenetic imprinting of a region of chromosome 11 → overexpression of insulin-like growth factor-2 and loss of expression of CDKN1C, a tumor suppressor gene that encodes the cyclin-dependent kinase inhibitor p57

ONE LINERS

- Fibropolycystic liver disease often occurs in association with autosomal recessive polycystic renal disease. Persons with fibropolycystic liver disease are at increased risk for cholangiocarcinoma
- Rare causes of acute liver failure include abnormalities of blood flow, metabolic disorders, and malignancies, most commonly leukemia or lymphoma (33%), followed by breast cancer (30%) and colon cancer (7%).
- The receptor for HAV on hepatocytes is HAVcr-1 (also known as TIM-1), a glycoprotein that is a member of a family of proteins that serve as receptors for several other viruses as well.
- HAV is not cytopathic, and the hepatocellular injury is inflicted by cytotoxic T-lymphocytes and NK cells that recognize and kill hepatocytes infected by the virus.
- Entry of HBV into hepatocytes is enabled by binding of large HbsAg to the bile salt transporter known as sodium taurocholate cotransporting polypeptide (NTCP)
- The supersaturated cholesterol in bile that forms cholesterol stones may diffuse into the mucosa and manifest as cholesterolosis
- Resection is recommended for β-catenin–activated tumors as well as tumors 5 cm or larger due to the risk of hemorrhage and malignant transformation
- Loss-of-function mutations in the HNF1-α gene, which encodes a transcription factor that regulates many genes in hepatocytes and in pancreatic islets, one of the causes of hepatic adenoma
- The liver may also be secondarily involved by other infections during pregnancy. These include hepatitis caused by herpes simplex virus, a rare cause of acute liver failure in pregnancy, and liver abscess caused by Listeria monocytogenes, an organism that thrives in placental tissue, from where it may seed the liver
- Fibrolamellar HCC: One unusual histologic subtype that often occurs in adolescents and young adults in the absence of preexisting liver disease, fibrolamellar HCC, is strongly associated with a fusion gene that leads to aberrant activity of protein kinase A, an enzyme that participates in a signaling pathway regulated by cAMP.
- Adenocarcinomas arising from the intrahepatic biliary tree are referred to as intrahepatic cholangiocarcinoma, while similar tumor arising from the extrahepatic bile ducts are referred to as biliary adenocarcinoma.
- Intrahepatic cholangiocarcinoma is the most common primary malignant tumor of the liver after HCC.

- Microscopically, gallbladder carcinoma are usually characterized by the presence of glands embedded in desmoplastic stroma, but in some cases cytologic atypia and stromal response are minimal; in these instances, identification of perineural and vascular invasion help establish the diagnosis.
- Alagille syndrome is an autosomal dominant disorder associated with cholestasis and paucity of bile ducts as well as other abnormalities such as dysmorphic facies, butterfly-shaped vertebra, eye defects, and cardiac defects.
- Dubin-Johnson syndrome is caused by mutation in the MRP2 gene (multidrug resistant protein 2), which is required for transport of non–bile salt organic anions at the canalicular membranes. Deposition of brown-black melanin-like pigment in the hepatocytes is a striking feature in this disease and can lead to blackening of the liver.
- The hallmark on liver biopsies in hereditary hyperbilirubinemia is portal expansion due to edema, prominent ductular reaction at the portal-parenchymal interface, and infiltrating neutrophils associated with ductules ("pericholangitis")
- In ascending cholangitis, the neutrophils also involve the bile duct epithelium and lumens. Persistent obstruction leads to fibrosis, which can eventually proceed to biliary cirrhosis
- Levels of UGT1A1 (uridine diphosphate-glucuronyl transferase family, peptide A1), the enzyme that is responsible for bilirubin glucuronidation, are low at birth and do not reach adult levels until 3 to 4 months of age
- In most infants, phototherapy with blue light (which converts bilirubin to a soluble isomer that is readily excreted in the urine) is sufficient to keep the levels of unconjugated bilirubin within a safe range until the hepatic machinery for conjugation
- Erythroferrone released from an expanded population of erythroid progenitors in the bone marrow suppress hepcidin production by the liver, leading to increased iron absorption causing hemochromatosis
- Transporter proteins examples: MRP2 (multidrugresistant protein-2) for conjugated bilirubin, bile salt export pump (BSEP) for bile salts, MDR3 (multi-drug resistance-3) for phosphatidylcholine, and sterolins 1 and 2 for cholesterol
- Consequence of hepatocellular accumulation of the misfolded protein in ∝1-Antitrypsin Deficiency is an example of a "toxic gain-of-function" mutation
- Toll-like receptor 4. Toll-like receptor signaling triggers the release of cytokines and chemokines, which may contribute to the inflammation that accompanies alcoholic liver disease.

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